

MANUAL

M1020
PERMANENT MAGNET
MOTOR CONTROLLER

VERSION: 1.0

MAIKAI TECHNOLOGY

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1. OVERVIEW

The M1020 controller is a permanent magnet motor speed controller. It offers smooth, silent, cost effective control of motor speed and torque. The M1020 controller is designed for use in mobility aid scooters and other small electric vehicles, such as sweeper/scrubbers.

Features include:

- ✧ Full bridge power MOSFET design, providing infinitely variable forward, reverse, drive, and brake control
- ✧ Programmability through the M1902 handheld programmer
- ✧ Complete diagnostics through the M1902 handheld programmer and a status LED
- ✧ Available for single-ended or wigwag 5k Ω potentiometer throttles and 5V throttles
- ✧ Two different operating modes allowing optimization of vehicle characteristics for different driving conditions
- ✧ Speed limit input provides linear variable speed limiting
- ✧ Current limiting in both driving and regenerative braking modes
- ✧ Load compensation stabilizes speed on ramps and over obstacles
- ✧ Speed scaler eliminates variations in maximum speed that would otherwise result from variations in battery charge levels
- ✧ High pedal disable (HPD) function with lockout monitors status of the throttle during turn-on and prevents operation until the throttle has been returned to neutral
- ✧ Key-off deceleration function provides a controlled deceleration if the keyswitch is turned off while driving
- ✧ "Emergency Stop" provides a faster deceleration in emergency condition
- ✧ Anti-rollback/anti-roll-forward function sets brake delay according to speed and direction for improved braking response and minimized rollback on hills, etc.
- ✧ "Push" input releases brake for key-on pushing
- ✧ Missing/shorted brake detection forces neutral in the event of an open or shorted brake circuit
- ✧ Reverse beeper option can be used to drive a low current dc horn when vehicle is in reverse
- ✧ Inhibit input disables the controller and puts the vehicle in a safe state during charging, etc.

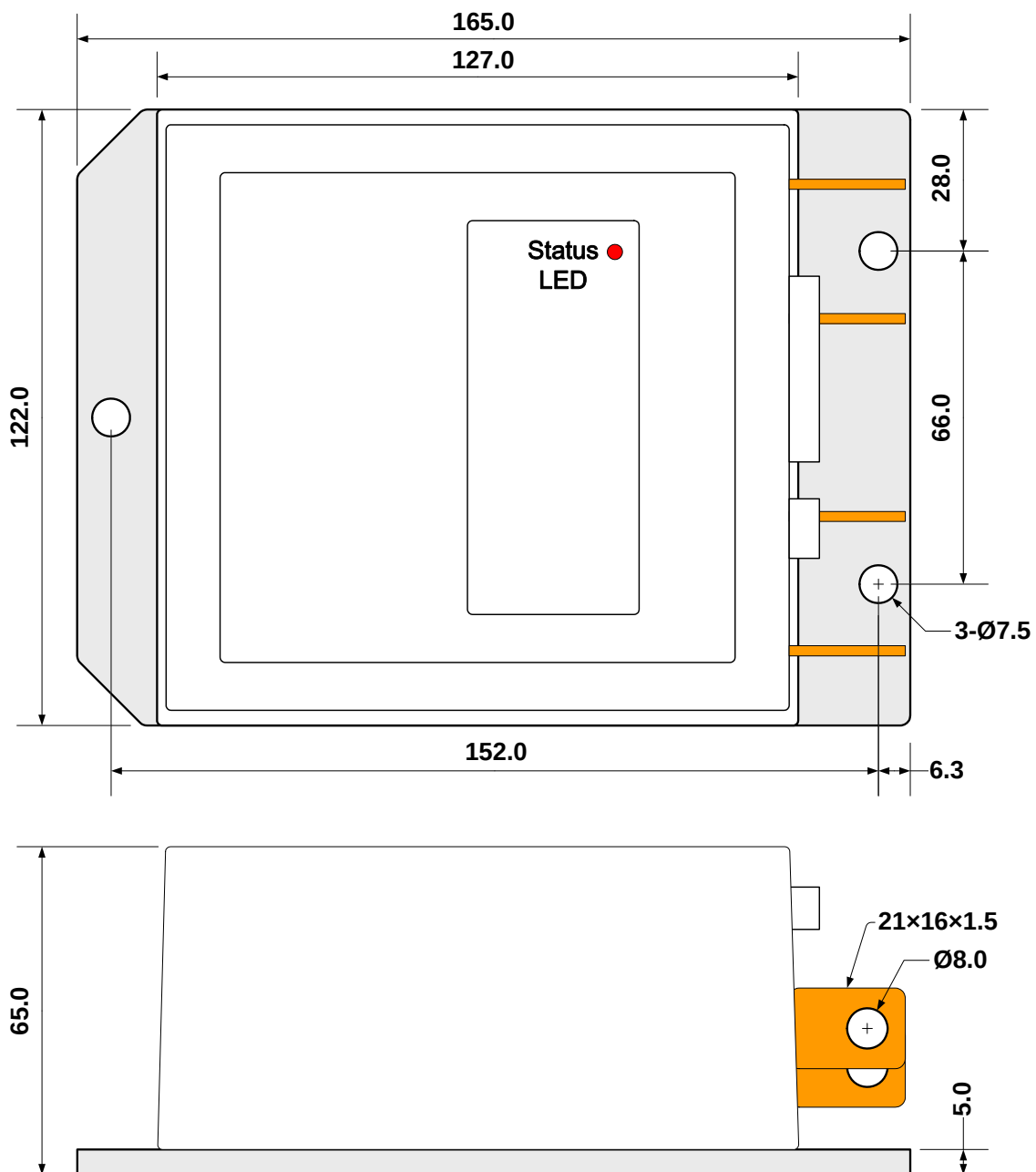
- ✧ Undervoltage cutback function protects against operation at low battery voltage
- ✧ Overvoltage protection shorts the motor and disables driving in the event of excessive battery voltage
- ✧ Thermally protected and compensated for stable output and overtemperature protection
- ✧ Programmable low side driver can be programmed to enable a side broom or brush contactor, hour meter, etc.
- ✧ Brake PWM allows the brake driver to be programmed to a reduced holding Voltage
- ✧ Reverse polarity protected (battery input)
- ✧ SRO input sequencing options
- ✧ Emergency reverse (belly button) input causes rapid transition to reverse
- ✧ Programmable high side driver output for brake light, hour meter, brush contactor, etc.
- ✧ ISO 7176 compliant throttle fault detection circuitry inhibits driving if throttle pot signal goes out of range for any reason

2. DIMENSIONS

The outline and dimensions are shown below.

To ensure full rated output power, the controller should be fastened to a clean, flat metal surface with three screws.

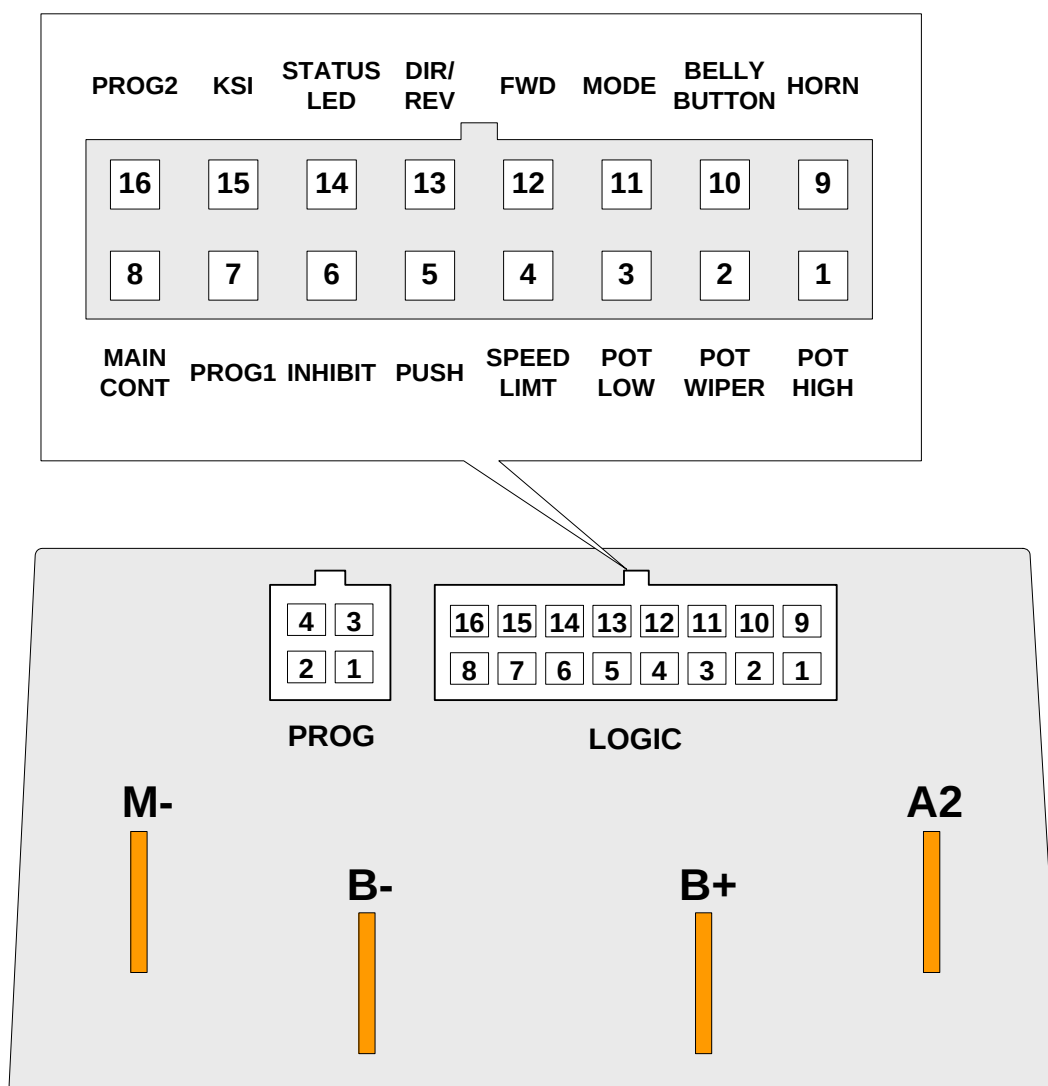
The controller can be mounted in any position, but the location should be carefully chosen to keep the controller clean and dry.



Dimensions in millimeters

3. INATALLATION & WIRING

3.1. Connectors Overview



3.2. High Current Connections

Four copper bus bars are provided for the high current connections to the battery (B+, B-) and motor (M-, A2).

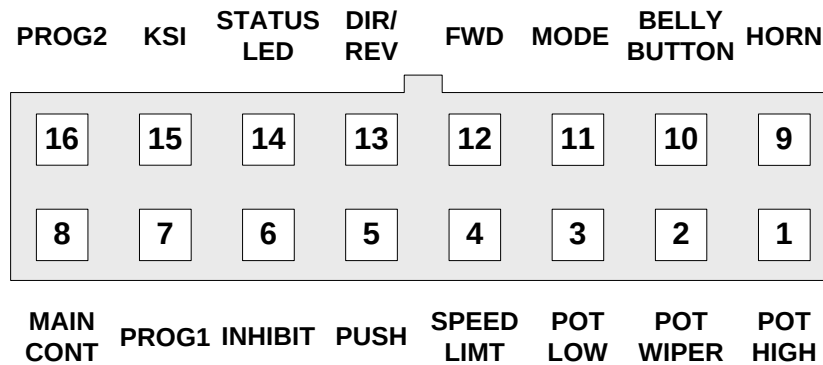
Cables are fastened to the bus bars by M8 bolts. When tightening the bolts, two opposing wrenches should be used to prevent bending the bus bars and putting undue strain on the internal connections.

3.3. Programmer Connections

A 4-pin low power connector is provided for the programmer.

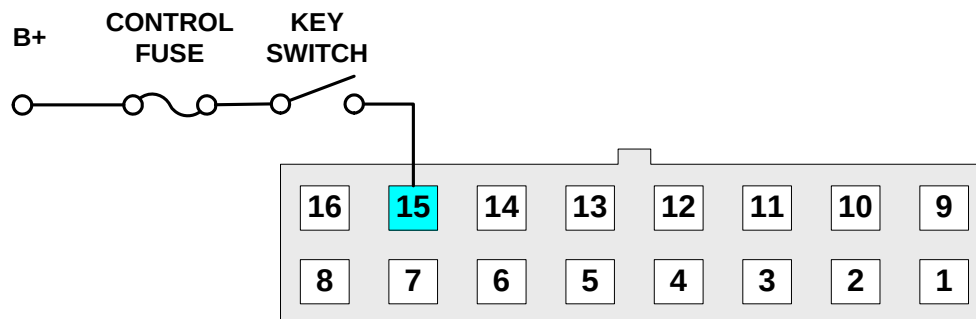
3.4. Low Current Connections

The low current logic control connections are provided by an 16-pin connector. The Molex Mini-Fit Jr. p/n 39-01-2165 with type 5556 pins is the mating connector.



3.4.1. Key Switch

The vehicle should have an OEM-supplied master on/off switch to turn the system off when not in use. The key switch provides logic power for the controller and for the other control input switches.



3.4.2. Speed Limit

A speed limit pot allows the operator to adjust the speed of the vehicle at full throttle.

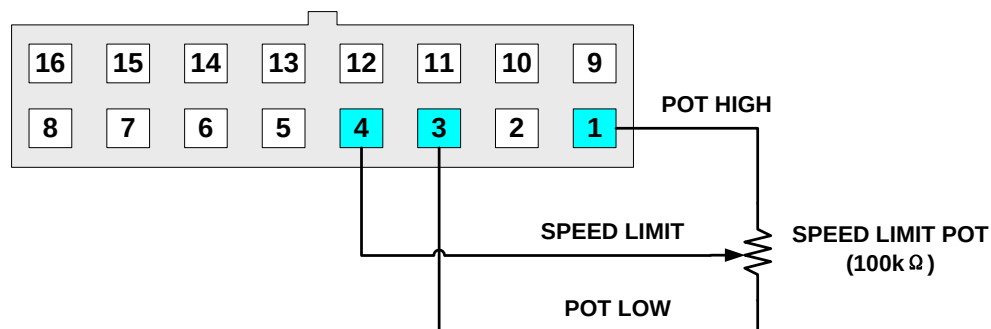
The speed limit pot should be sized so that it does not affect throttle input resistance and thus the throttle response. A 100k Ω pot is recommended.

The speed limit pot is at its maximum speed setting when its wiper is shorted to the throttle pot's pot high connection (Pin 1). When the speed limit pot is in its maximum speed position, the vehicle's speed at full throttle corresponds to the programmed maximum speed setting.

The speed limit pot is at its minimum speed setting when its wiper is shorted to the throttle pot's pot low connection (Pin 3). When the speed limit pot is in its minimum speed position, the vehicle's speed at full throttle corresponds to the programmed minimum speed setting.

If a speed limit pot is not used, the speed limit input (Pin 4) should be jumpered to the pot high output (Pin 1). In this configuration, the vehicle speed at full throttle is defined by the programmed maximum speed. If no jumper is used, the vehicle speed at full throttle will be limited to the programmed minimum speed.

If a speed limit pot will never be used in the application, the controller can be factory-configured without the speed limit feature, and the jumper will not be required. With such a controller, the vehicle speed at full throttle is defined by the programmed maximum speed.



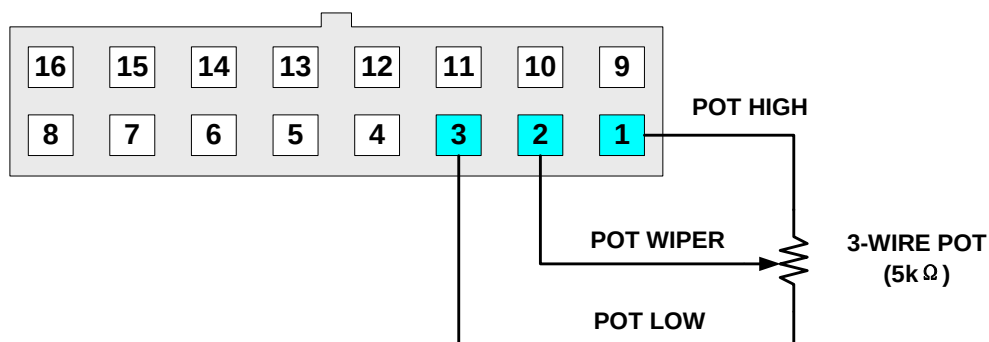
3.4.3. Throttle

A 3-wire potentiometer throttle or a voltage throttle can be used. The controller can accept a single-ended, inverse single-ended, wigwag, or inverse wigwag input signal from the throttle, depending on how the **THROTTLE TYPE** parameter is programmed.

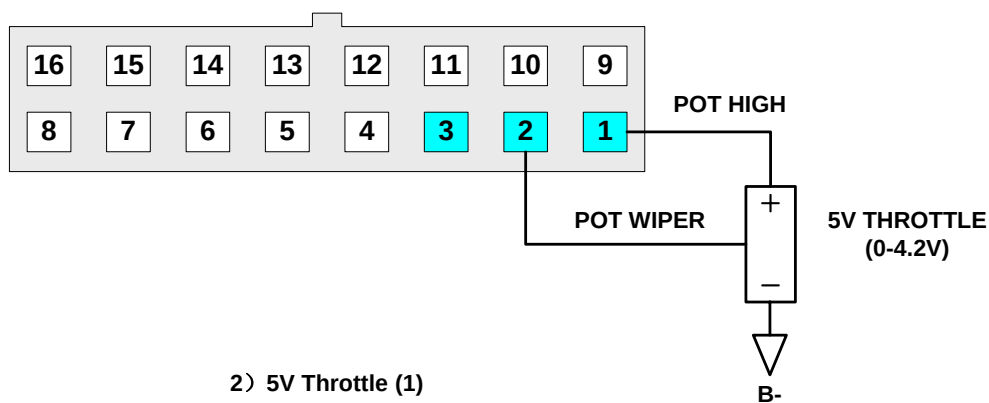
Throttle types are:

0-5K, 0-2.5-5K, 0-5V, 0-2.5-5V, 5-0K, 5-2.5-0K, 5-0V, 5-2.5-0V

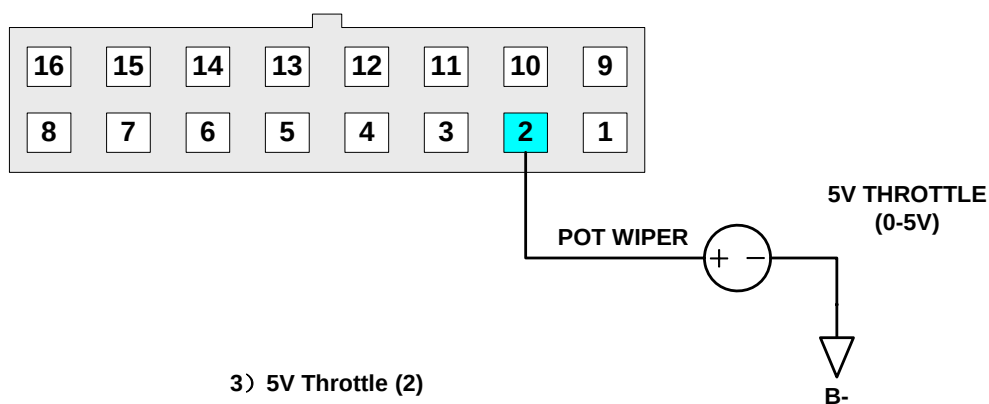
- ✧ potentiometer throttle : 0-5K, 0-2. 5-5K, 5-0K, 5-2. 5-0K
- ✧ voltage throttle: 0-5V, 0-2. 5-5V, 5-0V, 5-2. 5-0V
- ✧ direct proportion throttle : 0-5K, 0-2. 5-5K, 0-5V, 0-2. 5-5V
- ✧ inverse proportion throttle : 5-0K, 5-2. 5-0K, 5-0V, 5-2. 5-0V
- ✧ single-ended throttle : 0-5K, 0-5V, 5-0K, 5-0V
- ✧ wigwag throttle : 0-2. 5-5K, 0-2. 5-5V, 5-2. 5-0K, 5-2. 5-0V



1) 5kΩ 3-Wire Potentiometer



2) 5V Throttle (1)



3) 5V Throttle (2)

3.4.4. Direction

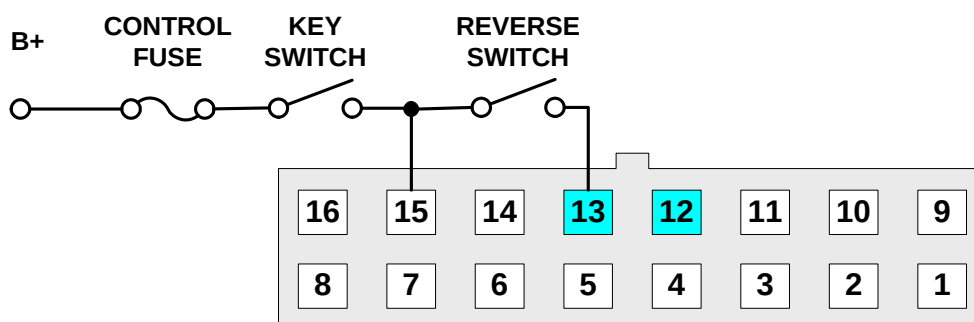
The **DIRECTION SWITCHES** parameter defines how the controller decides when to change direction.

If the parameter is set to “ 0 ”, a wigwag throttle existed, the direction input type defaults to respond to the throttle input.

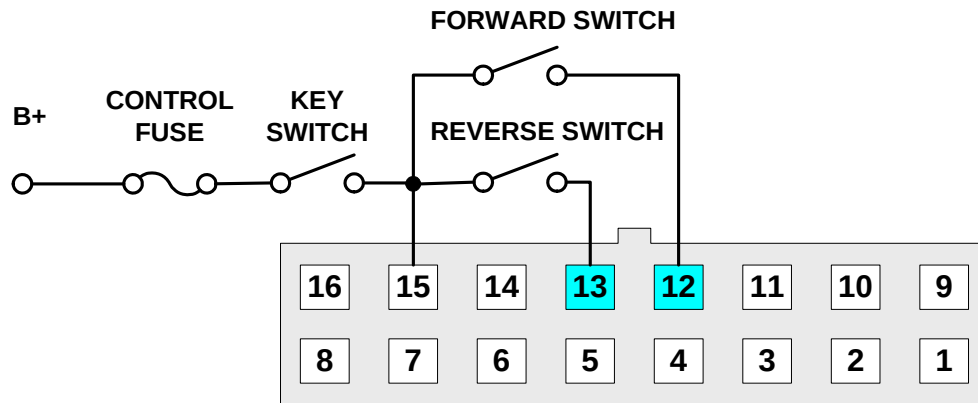
If the parameter is set to “ 1 ”, a single-ended throttle existed, the direction is chosen via a single switch, which pulls the direction/reverse input (Pin 13) high when reverse is selected. It is not necessary for the switch to connect Pin 13 to B- when forward is selected.

If the parameter is set to “ 2 ”, a single-ended throttle existed, there is two switches: one for reverse (Pin 13) and one for forward (Pin 12). the switch must pull the forward or reverse input to B+ to select the direction. Selecting both forward and reverse at the same time will cause the controller to inhibit the output, disabling the vehicle until only one direction is selected.

DIRECTION SWITCHES	THROTTLE TYPE	DESCRIPTION
0	wigwag	No switch, responds to throttle input
1	Single-Ended	Single Switch, allows direction to be selected via a single switch
2	Single-Ended	Dual Switches, requires direction to be selected via inputs from individual forward and reverse switches



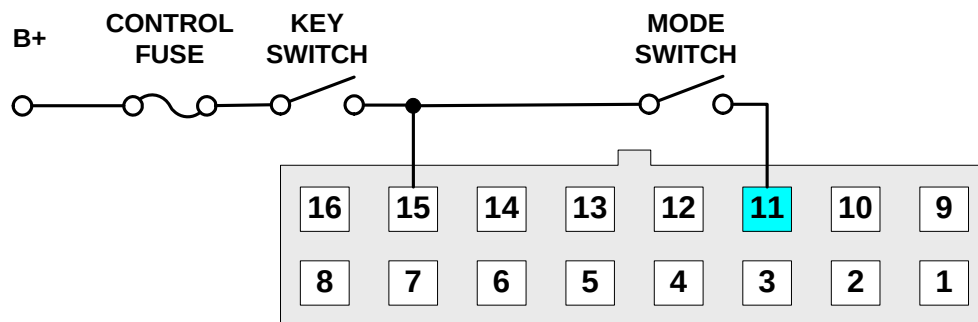
1) One Direction Switch



2) Two Direction Switches

3.4.5. Mode

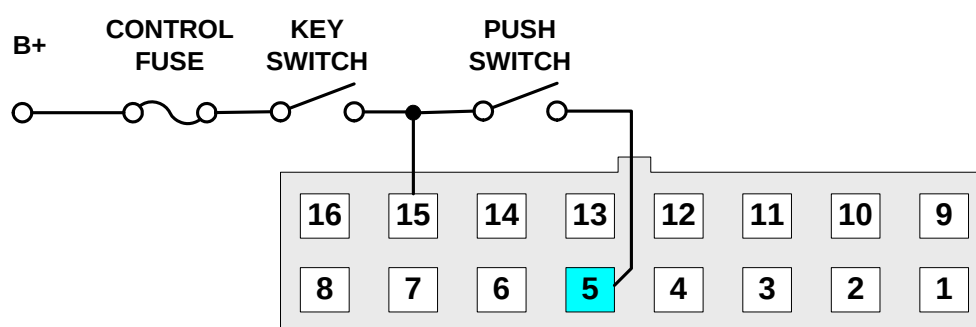
The controller allows operation in two distinct modes: “Mode 1” and “Mode 2.” These modes can be programmed to provide two different sets of operating characteristics, which can be useful for operation in different conditions. For example, Mode 1 could be programmed such that the vehicle moves slowly for precise, indoor maneuvering and Mode 2 programmed for higher speed, long distance travel outdoors.



3.4.6. Push

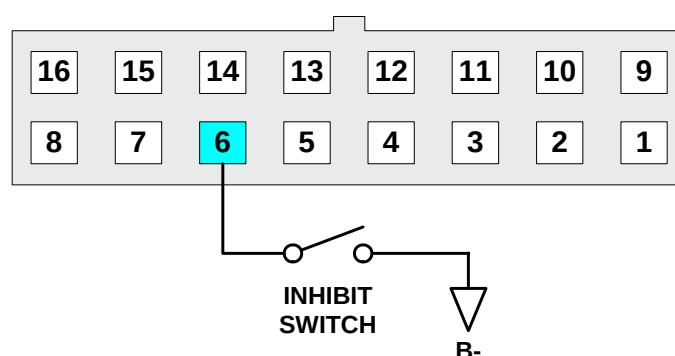
A push switch can be used to electrically release the electromagnetic brake, so that the vehicle can be pushed. Activating the push input inhibits the controller’s drive functions until the push switch is turned off.

The push switch must go from off to on while the vehicle is stopped; if the push switch is turned on while the vehicle is moving, the electromagnetic brake will not release when the vehicle stops. Also, the controller must be connected to the batteries and the keyswitch must be turned on in order for the push feature to be used.



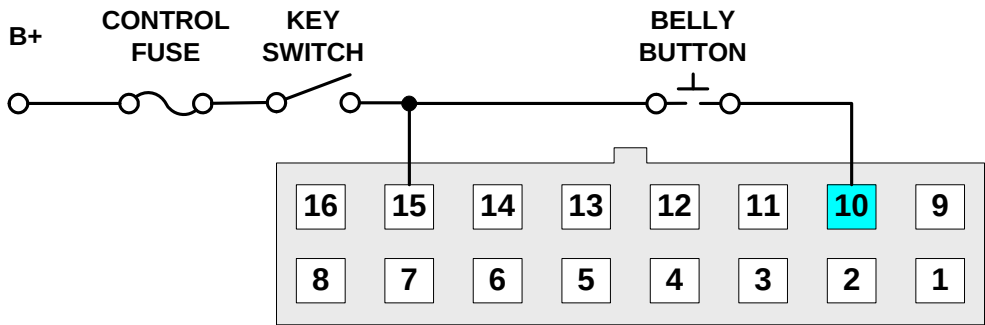
3.4.7. Inhibit

The inhibit input can be used to inhibit operation during battery charging. The inhibit input overrides all other controller inputs and is active when low (i.e., when shorted to B-). The input can be left floating when not engaged; it does not need to be pulled high. Typically, battery chargers have a dedicated third terminal that automatically provides inhibit.



3.4.8. Emergency Reverse

Emergency reverse is activated when the keyswitch is On and the emergency reverse switch (the BB, or “belly button” switch) is pressed. After the BB switch is released, normal controller operation is not resumed until zero throttle is selected. However, repeatedly pressing the BB switch will reactivate the emergency reverse function each time. The emergency reverse speed and current limit are both programmable.

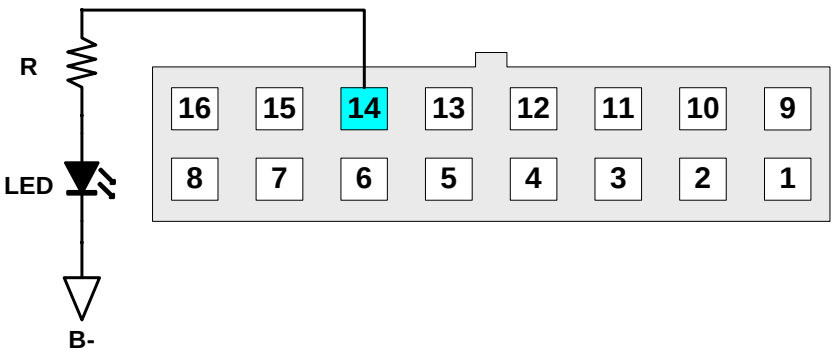


3.4.9. Status LED

The controller has the capability to drive a panel indicator LED, which can be used to tell the operator, at a glance, the controller’s status. This LED always indicates whether the controller is powered on or off. The status LED will also provide diagnostics information via flash codes.

If a status LED is used, it should be installed with the proper resistor in series. The controller’s LED driver is capable of a maximum current of 30 mA. The recommended resistors — designed to limit driver current to 15 mA when active — are listed bellow.

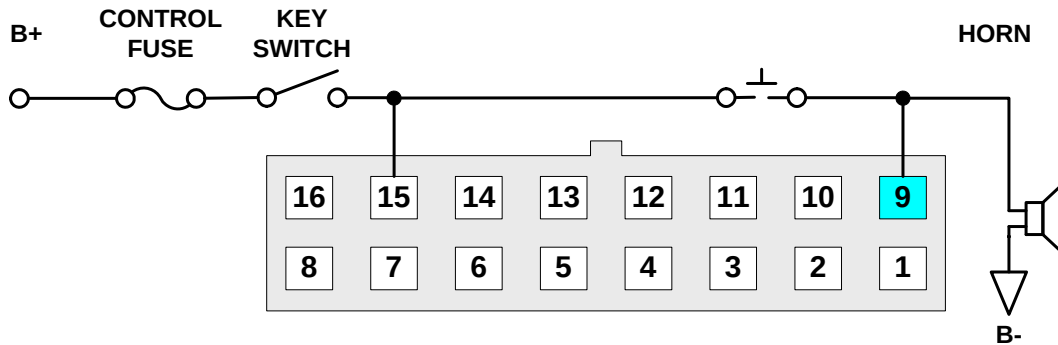
VOLTAGE	RESISTOR VALUE	POWER RATING
24V	1.5kΩ	0.5W
36V	2.4kΩ	1W
48V	3.3kΩ	2W



3.4.10. Horn

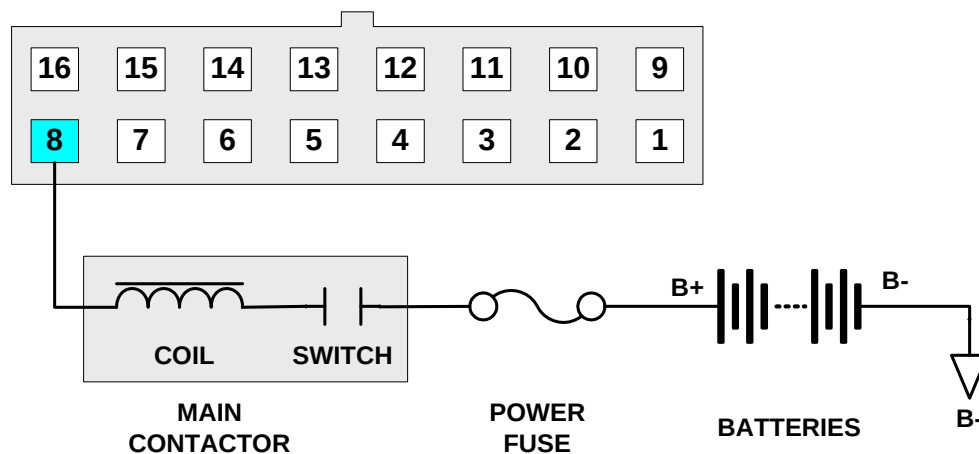
The controller’s horn driver(Pin 9) is designed to drive a low current dc horn at 1 Hz.

The horn sounds a warning when the reverse direction is selected (a series of beep tones). The horn driver provides a maximum current of 30 mA. Using a horn with a higher current requirement will damage and disable the driver.



3.4.11. Main Contactor

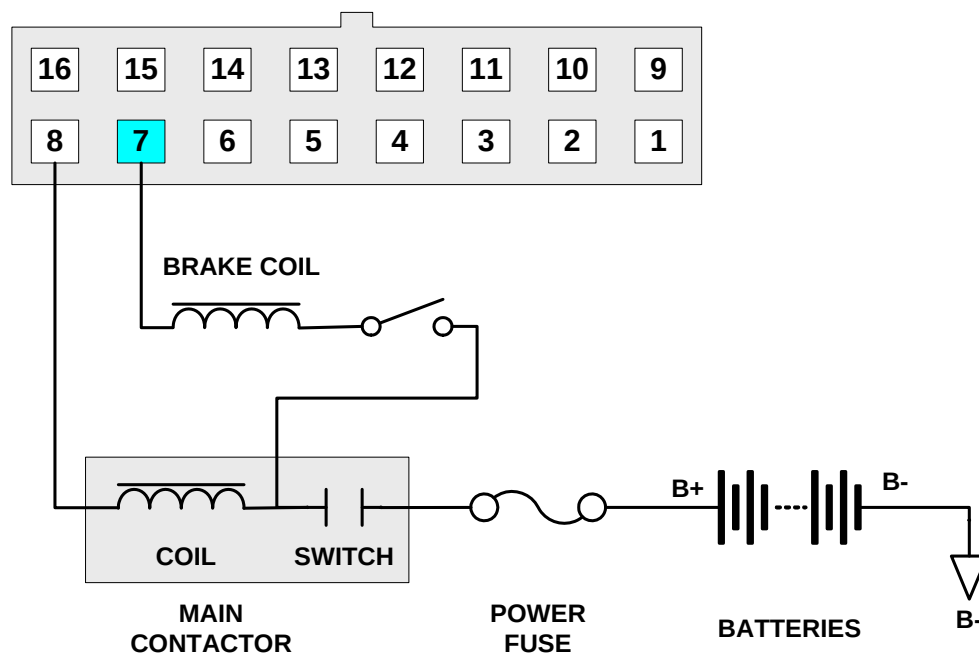
An external main contactor is required with the controller. A heavy-duty single-pole, single-throw (SPST) contactor with silver-alloy contacts is recommended. This contactor does not require an external precharge resistor, because of the controllers' built-in precharging feature.



3.4.12. Programmable Output 1/Brake

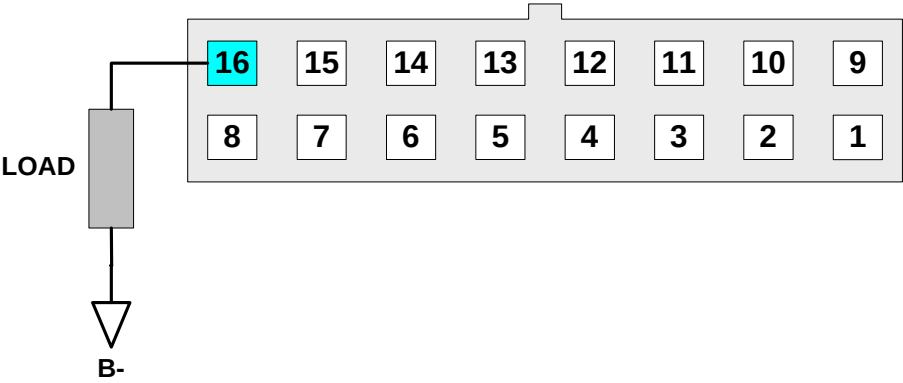
The program 1 type parameter configures a low side driver output at Pin 7. This driver is typically programmed to drive an electromagnetic brake. Alternatively, the program 1 driver can be programmed to drive an hour meter, sweeper/scrubber brush motor contactor, brake light, etc. It is rated at 1 ampere and is protected from external shorts.

When this output driver is programmed to drive an electromagnetic brake, if a brake release lever is used to release the electromagnetic brake mechanically, a brake coil disable switch is recommended. This switch opens the electromagnetic brake coil circuit when the mechanical brake release lever releases the brake from the motor shaft. The open brake coil circuit will register as a fault, inhibiting controller operation if an operator attempts to drive the vehicle with the brake mechanically released. This safety feature ensures that the vehicle cannot be driven when the brake cannot be engaged.

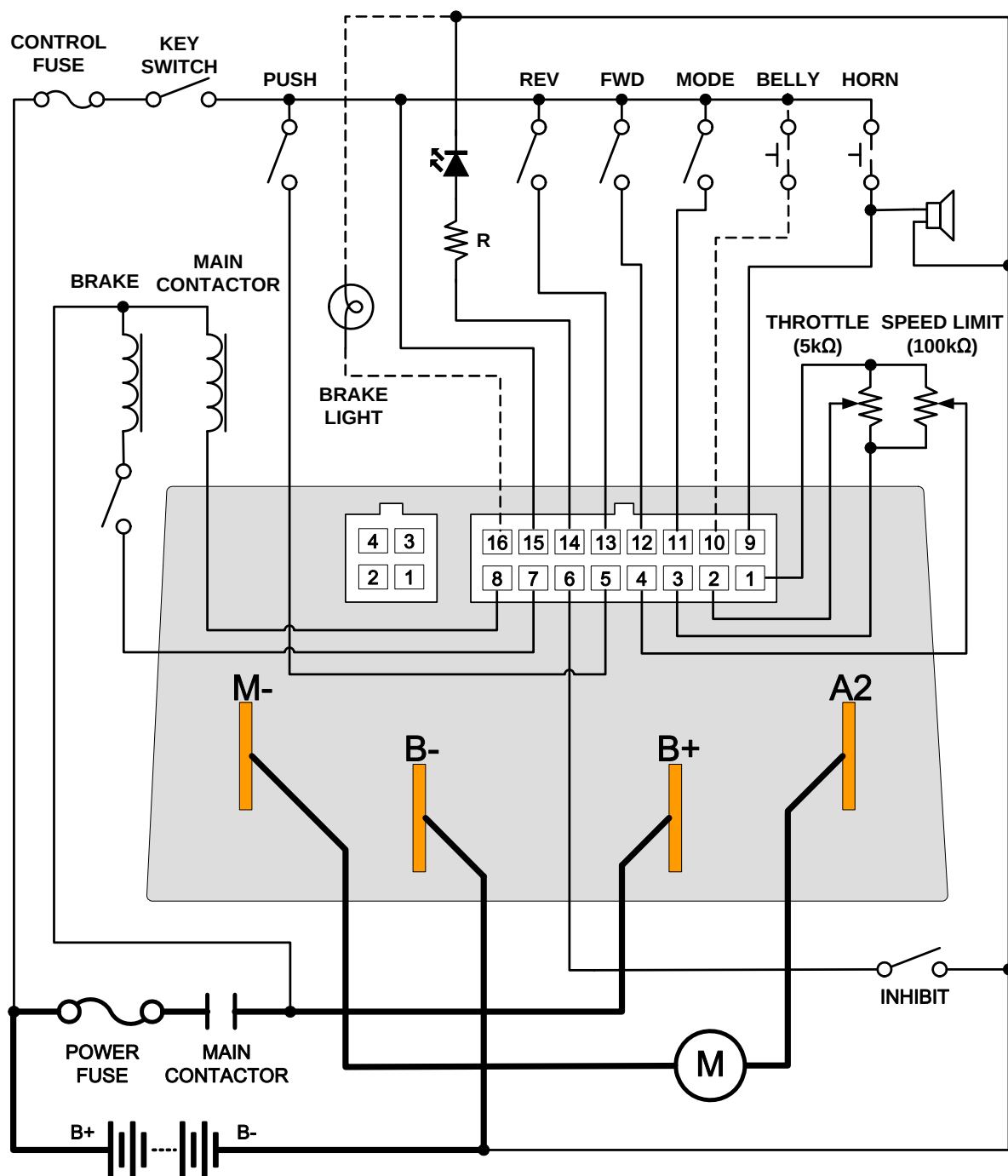


3.4.13. Programmable Output 2

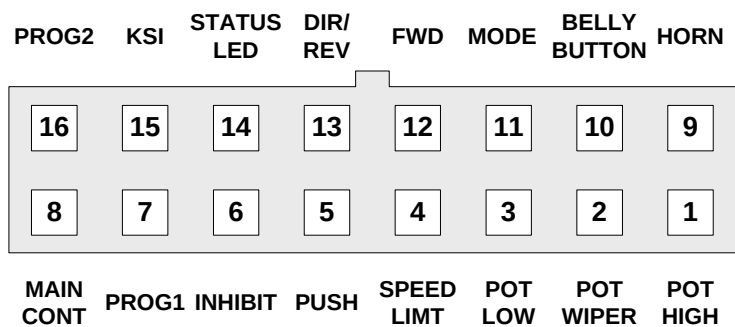
The program 2 type parameter configures a high side driver output at Pin 16. This driver can be programmed to drive an hour meter, belly button check output, sweeper/scrubber brush motor contactor, brake light, etc. It can also be configured to perform a BB wiring check. The program 2 driver is rated at 2 amperes and is not short circuit protected.



3.5. Wiring Example



Voltage	R
24V	1.5kΩ/0.5W
36V	2.4kΩ/1W
48V	3.3kΩ/2W



4. PROGRAMMABLE PARAMETERS

4.1. Parameters List

Parameters		Values of Programmer			Unit
		Minimum	Default	Maximum	
MOTOR PARAM					
	M1 MAIN C/L	50	50	100	percent
	M2 MAIN C/L	50	50	100	percent
	EMG REV C/L	20	50	100	percent
	REGEN C/L BOOST	0	0	25	percent
	OVER CURR DELAY	1	10	15	second
ACCEL PARAM					
	M1 FWD ACCEL	0.2	3.0	4.0	second
	M2 FWD ACCEL	0.2	3.0	4.0	second
	M1 REV ACCEL	0.2	3.0	4.0	second
	M2 REV ACCEL	0.2	3.0	4.0	second
	GEAR SOFTEN	0	50	100	percent
	SOFT START	0	50	100	percent
DECEL PARAM					
	M1 FWD DECEL	0.2	3.0	4.0	second
	M2 FWD DECEL	0.2	3.0	4.0	second
	M1 REV DECEL	0.2	3.0	4.0	second
	M2 REV DECEL	0.2	3.0	4.0	second
	EMG STOP DECEL	0.2	2.0	4.0	second
	KEY OFF DECEL	0.2	2.0	4.0	second
SPEED PARAM					
	M1 FWD MAX SPD	15	50	100	percent
	M1 FWD MIN SPD	0	10	60	percent
	M1 REV MAX SPD	15	50	100	percent
	M1 REV MIN SPD	0	10	60	percent
	M2 FWD MAX SPD	15	50	100	percent
	M2 FWD MIN SPD	0	10	60	percent
	M2 REV MAX SPD	15	50	100	percent
	M2 REV MIN SPD	0	10	60	percent
	CREEP SPD	0	8	25	percent
	PUSH SPD	25	25	50	percent

	MAX SPD VOLTS	80	120	120	percent
	EMG REV SPD	0	50	100	percent
	CON SPD ESCAPE	1	3	20	second
THRTL PARAM					
	THRTL TYPE	0-5K	0-5K	5-2.5-0V	
	DIR SWITCHES	0	1	2	
	THRTL DEADBAND	5	20	30	percent
	THRTL RANGE	50	80	95	percent
	RAMP SHAPE	20	50	70	percent
OUTPUT PARAM					
	PROG1 TYPE	NOT USED	NOT USED	TIMER1	
	PROG2 TYPE	NOT USED	NOT USED	BB CHECK	
	BRK HOLD VOLTS	0	0	100	percent
FAULT PARAM					
	HIGH PEDAL CHECK	NO	YES	YES	Yes/No
	SRO CHECK	0	0	3	
	FAULT BEEP	NO	NO	YES	Yes/No
OTHER PARAM					
	M1 IR COEFF	0	0	900	
	M2 IR COEFF	0	0	900	
	IR GRADE	1	16	32	
	BEEPER ALTER	NO	NO	YES	Yes/No
	SLEEP DELAY	0	25	60	minute
	MAIN CONT DELAY	0	25	60	second

4.2. Motor Parameters

4.2.1. M1/M2 MAIN C/L

The **MAIN CURRENT LIMIT** parameter allows adjustment of the maximum current the controller will supply to the motor during both drive and regen operation. This parameter can be limited to protect the motor from excessive (potentially damaging) currents or to reduce the maximum torque applied to the drive system by the motor. It is adjustable from 50% to 100% of the rated maximum controller current.

4.2.2. EMG REV C/L

The **EMERGENCY REVERSE CURRENT LIMIT** parameter defines the controller's current limit, regardless of the selected mode, when the belly button input (Pin 10) is active. The emergency reverse current limit is adjustable from 0% to the full rated controller current, regardless of the M1 and M2 main current limit settings.

NOTE: If the controller is not factory-configured for the emergency reverse option, this parameter will not be displayed in the Program Menu.

4.2.3. REGEN C/L BOOST

The **REGEN CURRENT LIMIT BOOST** parameter allows the regenerative braking current limit to be set higher than the main current limit. This feature can provide higher braking torque to stop heavy vehicles on steep inclines. The regen current boost parameter is adjustable from 0–25% above the programmed main current limit.

4.2.4. OVER CURR DELAY

The **OVER CURRENT DELAY** parameter defines the time it takes the controller to protect after the current over MAIN CURRENT LIMIT, This time delay is adjustable from 1.0 to 15.0 seconds.

4.3. Acceleration Parameters

4.3.1. M1/M2 FWD ACCEL

The **FORWARD ACCELERATION RATE** defines the time it takes the controller to accelerate from 0% output to 100% output during forward travel.

Larger values represent a longer acceleration time and gentler starts, while smaller values represent faster acceleration.

The accel rate is adjustable from 0.2 seconds to 4.0 seconds. The recommended range is 0.5–3.0 seconds, with accel rates less than 0.5 seconds allowed for abrupt acceleration under special circumstances.

4.3.2. M1/M2 REV ACCEL

The **REVERSE ACCELERATION RATE** defines the time it takes the controller to accelerate from 0% output to 100% output during reverse travel.

Larger values represent a longer acceleration time and gentler starts, while smaller values represent faster acceleration.

The accel rate is adjustable from 0.2 seconds to 4.0 seconds. The recommended range is 0.5–3.0 seconds, with accel rates less than 0.5 seconds allowed for abrupt acceleration under special circumstances.

4.3.3. GEAR SOFTEN

The gear soften feature allows smooth pickup of gear slack in the transmission when torque is reversed; it affects all accelerations except those from zero speed. The effect of this feature is most noticeable when reapplying the throttle from neutral after decelerating from high speed but before coming to a stop. (See soft start parameter, below, for softening torque endpoints for accelerations from a complete stop.)

The **GEAR SOFTEN** parameter is adjustable from 0% to 100%, with 100% providing a great deal of softening and 0% eliminating the feature. The tradeoff in increasing the gear soften value is that acceleration response may be slowed somewhat, especially at higher values.

4.3.4. SOFT START

The soft start feature allows softened torque endpoints for forward/reverse accelerations from a complete stop. When accelerating from a stop, some users prefer the softened gear slack transitions this parameter can provide, while others prefer the vehicle to respond instantly.

The **SOFT START** parameter is adjustable from 0% to 100%, with 100% providing a great deal of softening and 0% eliminating the feature. The tradeoff in increasing the soft start value is that acceleration response may be slowed somewhat, especially at higher values.

4.4. Deceleration Parameters

4.4.1. M1/M2 FWD DECEL

The **FORWARD DECELERATION RATE** defines the time it takes the controller to decelerate from 100% output to 0% output when traveling in the forward direction.

A larger value represents a longer deceleration time and a gentler stop. Reducing the decel rate will reduce the stopping distance required. The decel rate should be set at a value that will ensure the vehicle stops within a safe distance when traveling at maximum speed. (NOTE: The maximum allowed vehicle stopping distance may be defined by local regulations.)

The decel rate is adjustable from 0.2 seconds to 4.0 seconds. The recommended range is 0.5–3.0 seconds, with decel rates less than 0.5 seconds allowed for abrupt stops under special circumstances.

4.4.2. M1/M2 REV DECEL

The **REVERSE DECELERATION RATE** defines the time it takes the controller to decelerate from maximum reverse speed to 0% output when traveling in the reverse direction.

A larger value represents a longer deceleration time and a gentler stop. Reducing the reverse decel rate will reduce the stopping distance required. The reverse decel rate should be set at a value that will ensure the vehicle stops within a safe distance when traveling at maximum reverse speed. (NOTE: The maximum allowed vehicle stopping distance may be defined by local regulations.) Lower values may be required to ensure rapid stops when driving down ramps in reverse.

The reverse decel rate is adjustable from 0.2 seconds to 4.0 seconds. The recommended range is 0.5–3.0 seconds, with reverse decel rates less than 0.5 seconds allowed for abrupt stops under special circumstances.

4.4.3. EMG STOP DECEL

The **EMERGENCY STOP DECELERATION RATE** defines the time it takes the vehicle to stop when a reverse throttle command >80% is given while the vehicle is moving forward. This gives the operator a way to stop more quickly when unexpected conditions arise.

When the EMG Stop feature is invoked the EMG Stop deceleration rate becomes the new forward deceleration rate. Therefore it makes sense to set it to a value lower (faster stop) than the fastest forward deceleration rate (DECEL MAX SPEED). The EMG Stop deceleration rate is adjustable from 0.2 to 4.0 seconds.

4.4.4. KEY OFF DECEL

The **KEY-OFF DECELERATION RATE** defines the time it takes the vehicle to stop after the key switch has been turned off while the vehicle is in motion. The key-off deceleration rate is independent of the normal programmed deceleration rate, the selected mode, and the speed and direction of travel when KSI (key switch input) is switched off. It is adjustable from 0.2 to 4.0 seconds.

4.5. Speed Parameters

4.5.1. M1/M2 FWD MAX SPD

The **MAXIMUM FORWARD SPEED** parameter defines the maximum allowed speed in forward at full forward throttle with the speed limit pot in its maximum speed position.

For example, if Mode 1 Maximum Speed is set at 60% and the speed limit pot is in its maximum speed position, the controller will adjust its output to achieve 60% speed at full throttle in Mode 1.

4.5.2. M1/M2 FWD MIN SPD

The **MINIMUM FORWARD SPEED** parameter defines the maximum allowed speed in forward at full forward throttle with the speed limit pot in its minimum speed position.

For example, if Mode 1 Minimum Speed is set at 20% and the speed limit pot is in its minimum speed position, the controller will adjust its output to achieve 20% speed at full throttle in Mode 1.

The minimum forward speed cannot be set higher than the programmed maximum forward speed.

4.5.3. M1/M2 REV MAX SPD

The **MAXIMUM REVERSE SPEED** parameter defines the maximum allowed speed in reverse at full throttle with the speed limit pot in its maximum speed position.

For example, if Mode 1 Maximum Reverse Speed is set at 40% and the speed limit pot is in its maximum speed position, the controller will adjust its output to achieve 40% reverse speed at full throttle in Mode 1.

4.5.4. M1/M2 REV MIN SPD

The **MINIMUM REVERSE SPEED** parameter defines the maximum allowed speed in reverse at full throttle with the speed limit pot in its minimum speed position.

For example, if Mode 1 Minimum Reverse Speed is set at 10% and the speed limit pot is in its minimum speed position, the controller will adjust its output to achieve 10% reverse speed at full throttle in Mode 1.

The minimum reverse speed cannot be set higher than the programmed maximum reverse speed.

4.5.5. CREEP SPD

CREEP SPEED helps to prevent vehicle rollback on inclines when the brake is released with very little throttle applied. It is activated when the throttle request exceeds the throttle's deadband threshold. The throttle response is rescaled so that the controller's output is adjustable over the full throttle range, but starting at the programmed creep speed value.

Creep speed is programmable from 0% to 25.0% of the maximum available speed.

4.5.6. PUSH SPD

When the push switch is switched to the On position, the push feature releases the electromagnetic brake and allows the vehicle to be manually pushed. The maximum speed at which the vehicle can be pushed is defined by the push speed parameter. It is programmable from 25% to 50% of the maximum available speed.

4.5.7. MAX SPD VOLTS

The **MAXIMUM SPEED VOLTAGES** parameter sets the maximum voltage that can be applied to the motor. It can be used to eliminate variations in maximum speed that would otherwise result when driving with a fully charged battery vs. a partially discharged battery.

The **MAX SPD VOLTS** parameter is a percent of Normal Battery Voltage, it programmable from 80% to 120%.

4.5.8. EMG REV SPD

The **EMERGENCY REVERSE SPEED** parameter sets the speed at which the vehicle moves away from the operator when the belly button input (Pin 10) is active. The vehicle will move at this speed any time the belly button input is active, regardless of throttle position. The emergency reverse speed is adjustable from 0% to 100% of the PWM duty cycle.

NOTE: If the controller is not factory configured for the emergency reverse option, this parameter will not be displayed in the Program Menu.

4.5.9. CON SPD ESCAPE

The **CONSTANT SPEED ESCAPE TIME** parameter defines the time it takes the controller to escape from constant speed operation to normal operation while there is an exception occur.

It programmable from 1-20 seconds.

4.6. Throttle Parameters

4.6.1. THRTL TYPE

A 3-wire potentiometer throttle or a voltage throttle can be used. The controller can accept a single-ended, inverse single-ended, wigwag, or inverse wigwag input signal from the throttle, depending on how the Throttle Type parameter is programmed.

throttle types are:

0-5K, 0-2. 5-5K, 0-5V, 0-2. 5-5V, 5-0K, 5-2. 5-0K, 5-0V, 5-2. 5-0V

- ✧ potentiometer throttle : 0-5K, 0-2. 5-5K, 5-0K, 5-2. 5-0K
- ✧ voltage throttle: 0-5V, 0-2. 5-5V, 5-0V, 5-2. 5-0V
- ✧ direct proportion throttle : 0-5K, 0-2. 5-5K, 0-5V, 0-2. 5-5V
- ✧ inverse proportion throttle : 5-0K, 5-2. 5-0K, 5-0V, 5-2. 5-0V
- ✧ single-ended throttle : 0-5K, 0-5V, 5-0K, 5-0V
- ✧ wigwag throttle : 0-2. 5-5K, 0-2. 5-5V, 5-2. 5-0K, 5-2. 5-0V

4.6.2. DIR SWITCHES

The **DIRECTION SWITCHES** defines how the controller decides when to change direction.

If the parameter is set to “ 0 ”, a wigwag throttle existed, the direction input type defaults to respond to the throttle input.

If the parameter is set to “ 1 ”, a single-ended throttle existed, the direction is chosen via a single switch, which pulls the direction/reverse input (Pin 13) high when reverse is selected. It is not necessary for the switch to connect Pin 13 to B- when forward is selected.

If the parameter is set to “ 2 ”, a single-ended throttle existed, there is two switches: one for reverse (Pin 13) and one for forward (Pin 12). the switch must pull the forward or reverse input to B+ to select the direction. Selecting both forward and reverse at the same time will cause the controller to inhibit the output, disabling the vehicle until only one direction is selected.

DIRECTION SWITCH NUMBER	THROTTLE TYPE	DESCRIPTION
0	wigwag	No switch, responds to throttle input
1	Single-Ended	Single Switch, allows direction to be selected via a single switch
2	Single-Ended	Dual Switches, requires direction to be selected via inputs from individual forward and reverse switches

4.6.3. THRTL DEADBAND

The **THROTTLE DEADBAND** parameter defines the throttle pot wiper voltage range that the controller interprets as neutral. Increasing the throttle deadband setting increases the neutral range. This parameter is especially useful with throttle assemblies that do not reliably return to a well-defined neutral point, because it allows the deadband to be defined wide enough to ensure that the controller goes into neutral when the throttle mechanism is released.

4.6.4. THRTL RANGE

The **THROTTLE RANGE** parameter sets the wiper voltage required to produce 100% controller output. Decreasing the throttle range setting reduces the wiper voltage required, and therefore the full stroke necessary to produce full output is reduced.

4.6.5. RAMP SHAPE

The **RAMP SHAPE** parameter determines the static throttle map of the controller. This parameter modifies the vehicle's response to the throttle input. Setting the ramp shape parameter at 50% provides a linear output response to throttle position. Values below 50% reduce the controller output at low throttle settings, providing enhanced slow speed maneuverability. Values above 50% give the vehicle a faster, jumpier feel at low throttle settings.

4.7. Fault Parameters

4.7.1. PROG1 TYPE

The **PROGRAM 1 TYPE** parameter configures an auxiliary low side driver (Pin 7). This output is rated at 1 amp and is short circuit protected. It can be programmed in any of eight configurations (Types 0–7 in the Program 1 Driver programming menu), as listed bellow.

Any component can be controlled by this output provided its current requirements do not exceed the driver's 1 amp rating.

TYPE	TYPICAL USE	DESCRIPTION OF OPERATION
NOT USED	Not used	Not used
BRAKE	Electromagnetic Brake Driver	Turns on when direction is selected* and throttle is applied. Turns off, after the specified brake delay, when PWM output returns to zero or as commanded by the anti-rollback and anti-roll-forward circuitry.
LIGHT	Brake Light Driver	Turns on when controller is in braking mode. Turns off, after a half-second delay, when braking stops.
CONTACT1	Brush Motor Contactor Coil Driver: Option 1	Turns on when direction is selected.* Turns off, without a delay, when PWM output returns to zero.
CONTACT2	Brush Motor Contactor Coil Driver: Option 2	Turns on when direction is selected.* Turns off, after a 2 s delay, when PWM output returns to zero.
CONTACT3	Brush Motor Contactor Coil Driver: Option 3	Turns on only when forward direction is selected.* Turns off, with no delay, in any other condition.
CONTACT4	Brush Motor Contactor Coil Driver: Option 4	Turns on only when forward direction is selected.* Turns off, after a 2 s delay, in any other condition.
TIMER1	Hour Meter	Turns on when power is turned on. Turns off when power is turned off.
* A direction is selected as follows: for wigwag throttles, when the throttle is rotated out of its neutral deadband; for single-ended throttles, when a direction input has been selected.		

4.7.2. PROG2 TYPE

The program 2 parameter configures an auxiliary high side driver at Pin 16. This output is rated at 2 amps and is not short circuit protected. It can be programmed in any of nine

configurations (Types 0–8 in the Program 2 Driver programming menu), as listed bellow.

Any component can be controlled by this output provided its current requirements do not exceed the driver's 2 amp rating. However, since this driver is not fault checked, it should not be used to drive an electromagnetic brake.

TYPE	TYPICAL USE	DESCRIPTION OF OPERATION
NOT USED	Not used	Not used
LIGHT	Brake Light Driver	Turns on when controller is in braking mode. Turns off, after a half-second delay, when braking stops.
CONTACT1	Brush Motor Contactor Coil Driver: Option 1	Turns on when direction is selected.* Turns off, without a delay, when PWM output returns to zero.
CONTACT2	Brush Motor Contactor Coil Driver: Option 2	Turns on when direction is selected.* Turns off, after a 2 s delay, when PWM output returns to zero.
CONTACT3	Brush Motor Contactor Coil Driver: Option 3	Turns on only when forward direction is selected.* Turns off, with no delay, in any other condition.
CONTACT4	Brush Motor Contactor Coil Driver: Option 4	Turns on only when forward direction is selected.* Turns off, after a 2 s delay, in any other condition.
TIMER1	Hour Meter: Option 1	Turns on when power is turned on. Turns off when power is turned off.
TIMER2	Hour Meter: Option 2	Turns on when electromagnetic brake is released and a direction is selected.*
BB CHECK	Belly Button Check	Periodically checks continuity of the wiring from the belly button switch to the controller's emergency reverse input.
* A direction is selected as follows: for wigwag throttles, when the throttle is rotated out of its neutral deadband; for single-ended throttles, when a direction input has been selected.		

4.7.3. BRK HOLD VOLTS

The **BRAKE HOLDING VOLTAGE** parameter allows selection of a voltage less than the full battery voltage to be applied to the brake coil when the brake is energized. This parameter is applicable only when the Program 1 driver is configured as an electromagnetic brake driver. The voltage is pulse width modulated and the setting represents the percentage of nominal battery voltage applied to the coil. Setting the parameter value to 100% allows the nominal battery voltage (e.g., 24V) to be applied to the coil whenever the battery voltage exceeds its nominal value. Setting the parameter value to 0% turns this feature off, causing the driver to apply the actual DC battery voltage to the brake coil when the brake is energized. When this parameter is set to a value greater than 0%, the nominal battery voltage is applied for 1 second (to ensure the brake coil disengages the brake) and then the programmed value is applied.

This feature provides greater vehicle range by reducing the power provided to the brake coil during operation. It also reduces brake heating, and thus reduces degradation of the brake's pull-in characteristics at high temperatures.

4.8. Fault Parameters

4.8.1. HIGH PEDAL CHECK

The primary function of the high pedal disable (HPD) feature is to prevent the vehicle from moving if the controller is turned on with the throttle already applied. HPD also serves as the interlock to prevent the vehicle from starting up with the push or inhibit feature active, and to prevent driving if Inhibit is activated during operation.

When the HPD parameter is programmed On, HPD is active and controller output is inhibited (1) if a throttle input greater than the throttle deadband exists when the controller is turned on, (2) if the push switch is On when the controller is turned on, (3) after the vehicle reaches a stop if the push switch is activated while the vehicle is being driven, or (4) if the inhibit switch is activated while the vehicle is being driven. If HPD is programmed Off, this protection feature is disabled.

Note: To meet TÜV requirements, the HPD feature must be programmed On.

4.8.2. SRO CHECK

The static return to off (SRO) feature prevents the vehicle from being started when “in gear.” If a direction is selected before the keyswitch and power enable switch (is one is used) are turned on, the SRO feature will inhibit the controller’s output.

TYPE	DESCRIPTION OF OPERATION
NO	Disable SRO function
FWD&REV	Check either direction
FWD	Check forward only
REV	Check Reverse only

4.8.3. FAULT BEEP

The fault beep parameter enables the horn during controller faults, in order to make the fault codes audible. It beeps only the fault codes; it does not precede the fault code with a level-of-seriousness code (as does the status LED, with its slow/fast flash preceding the fault code). If this audible alarm is not wanted, the fault beep parameter should be programmed Off.

4.9. Other Parameters

4.9.1. M1/M2 IR COEFF

IR compensation is a method by which the controller maintains a constant vehicle speed despite changes in motor loading. The IR speed coefficient parameter adjusts how aggressively the controller tries to maintain constant speed under changing load conditions. This parameter should be set at or below the resistance (in milliohms) of the traction drive motor system including motor wiring and connections. The IR speed coefficient should never be set higher than this value, because jerky and uncomfortable or unsafe operation could result. The IR speed coefficient is adjustable down to 0, with 0 equaling no IR compensation.

4.9.2. IR GRADE

The **IR STIFFNESS GRADE** parameter allows adjustment of the IR compensation characteristics at low speeds or light loads. The IR stiffness parameter can be programmed from 0 to 32. Increasing the IR stiffness value makes the IR speed coefficient response more aggressive at low speeds or light loads, but it also makes the throttle response more harsh and abrupt. The IR stiffness value should therefore be set to maximize the IR compensation response while maintaining as smooth a throttle response as possible. For most applications, the midpoint setting (16) is a good compromise.

4.9.3. BEEPER ALTER

The **BEEPER ALTER** parameter, when programmed Off, provides a continuous 24V+ signal to the horn driver (Pin 16) when throttle is requested in reverse; this signal can be used to drive a logic function—such as a watering solenoid for a sweeper/scrubber.

When a horn is connected to Pin 16, the Beeper alter parameter is typically programmed on. With Beeper alter programmed on, the horn sounds a series of beep tones when throttle is requested in reverse.

4.9.4. SLEEP DELAY

The controller powers down completely if the throttle request remains at neutral beyond the time specified by the sleep delay parameter; to resume operation, the key switch must be cycled. The sleep delay can be set from 0 to 60 minutes. Setting this parameter to zero disables the sleep delay.

4.9.5. MAIN CONT DELAY

The controller releases the main contactor if the throttle request remains at neutral beyond the time specified by the **MAIN CONTACTOR DELAY** parameter. The main contactor delay can be set from 0 to 60 seconds. Setting this parameter to zero disables the main contactor delay.

5. STATUS AND DISGNOTICS

LED CODES			EXPLANATION
		OFF	no power or defective controller
		SOLID ON	controller operational; no faults
		FLASH ONCE PER SECOND	push; no faults
		FLASH ONCE EACH 2 SECOND	inhibit; no faults
		FLASH ONCE EACH 5 SECOND	sleep; no faults
★	1, 1	○ ○	power section fault current protect
★	1, 2	○ ○○	main contactor fault precharge fault motor voltage fault (hardware failsafe)
▲	1, 3	○ ○○○	EEPROM fault
	2, 1	○○ ○	SRO fault
	2, 2	○○ ○○	HPD (High Pedal Disable) fault
★	2, 3	○○ ○○○	HPD fault present for >10 seconds
★	2, 4	○○ ○○○○	speed limit pot fault
	3, 1	○○○ ○	Emergency reverse check fault
★	3, 2	○○○ ○○	brake fault
★	3, 3	○○○ ○○○	throttle fault
	4, 1	○○○○ ○	undervoltage fault
	4, 2	○○○○ ○○	overvoltage fault
★	4, 3	○○○○ ○○○	thermal cutback fault
★	5, 5	○○○○○ ○○○○	controller software fault

★ *Must cycle key switch to clear.*

▲ *Must use programmer to clear, as follows: select Program Menu, alter data value of any parameter, cycle key switch.*

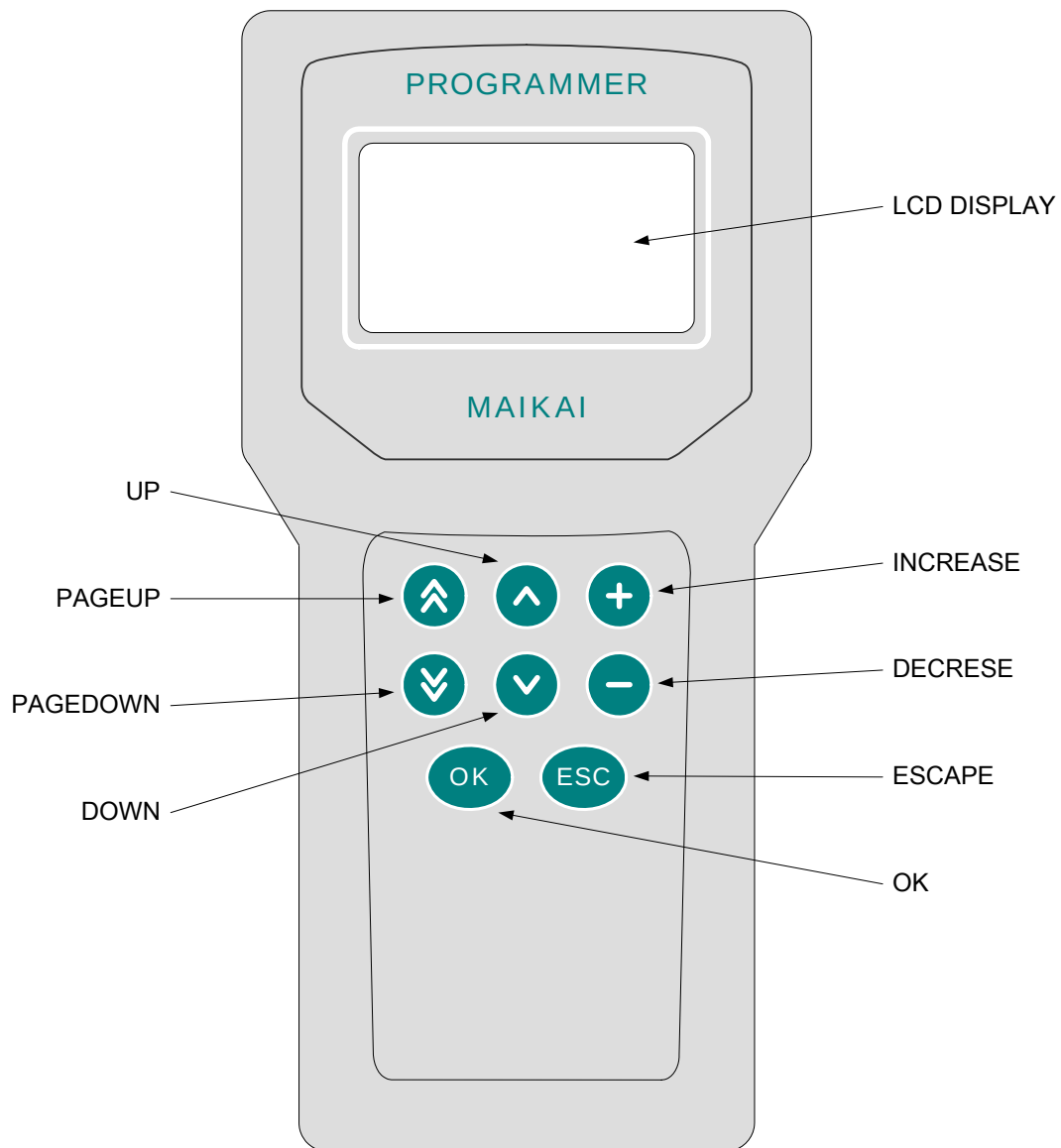
6. SPECIFICATIONS

Model	Normal Battery Voltage (Volts)	15 SEC Rating Current (Amps)	1 MIN Rating Current (Amps)	1 HOUR Rating Current (Amps)	Under- Voltage Cutback (Volts)	Over- Voltage Cutoff (Volts)
M1020-24V160A	24	160	160*	60	18	36
M1020-24V200A	24	200	200*	70	18	36
M1020-36V120A	36	120	120*	50	27	48
M1020-36V160A	36	160	160*	60	27	48
M1020-36V200A	36	200	200*	70	27	63
M1020-48V120A	48	120	120*	40	36	63

* Actual value of 1-minute rating depends on MOSFET heating.

APPENDIX A: M1902 HANDHELD PROGRAMMER

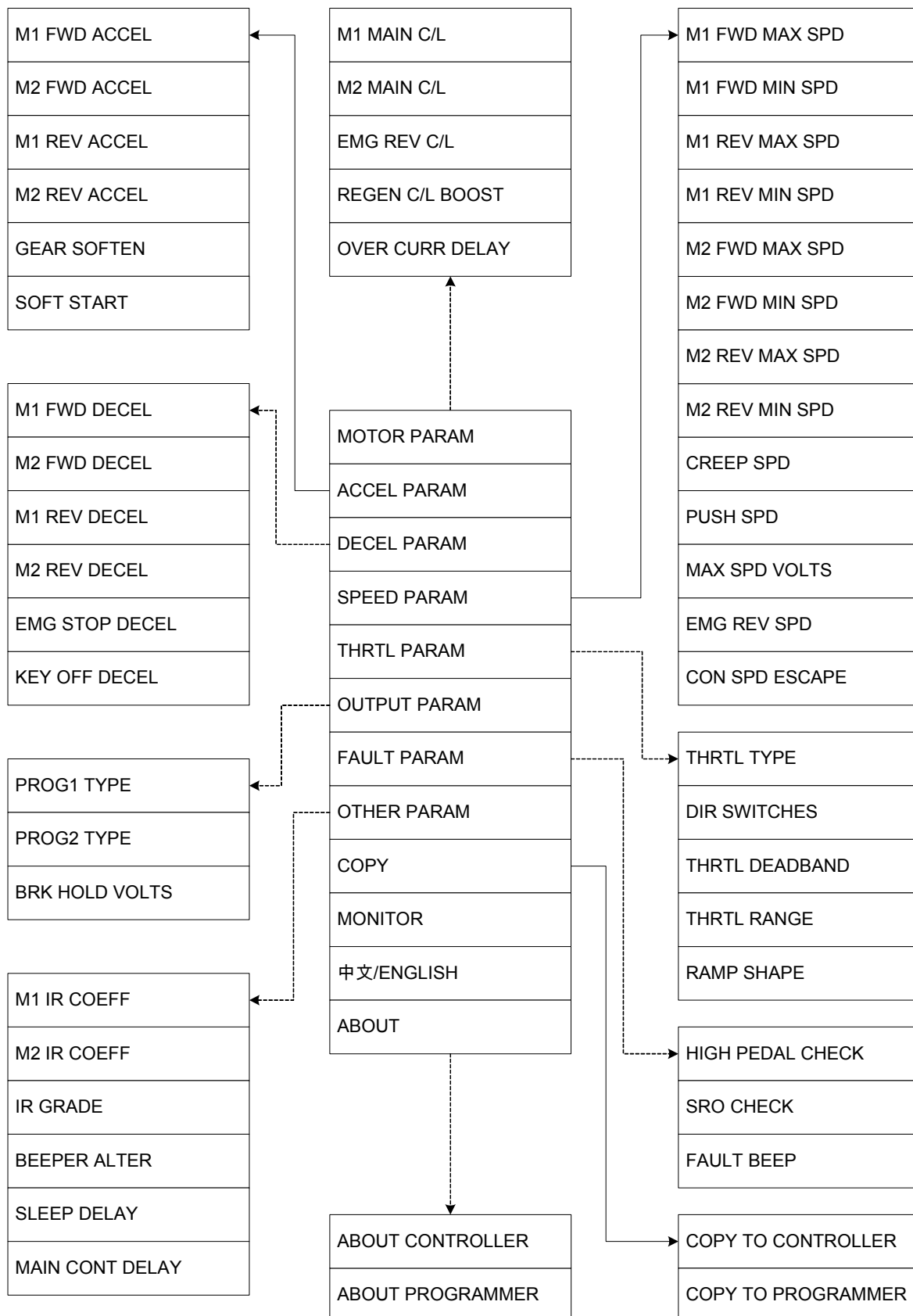
The M1902 handheld programmer provides programming, monitor, and copy capabilities for the controller. The power for operating the programmer is supplied by the host controller via the 4-pin connector.



A.1 Buttons

	PageUp Button
	PageDown Button
	Up Button
	Down Button
	Increase Button
	Decrease Button
	Enter to lower menu
	Escape to upper menu

A.2 Menus



■ Top Menu

Menu Name	DESCRIPTION
MOTOR PARAM	Motor Parameters
ACCEL PARAM	Acceleration Parameters
DECEL PARAM	Deceleration Parameters
SPEED PARAM	Speed Parameters
THRTL PARAM	Throttle Parameters
OUTPUT PARAM	Power Oupt Parameters
FAULT PARAM	Fault Parameters
OTHER PARAM	Other Parameters
COPY	copy parameters from programmer to controller, or copy parameters from controller to programmer
MONITOR	Monitor the status of controller
中文/ENGLISH	Select language
ABOUT	Display information about programmer and controller

■ Motor Parameters

M1 MAIN C/L
M2 MAIN C/L
EMG REV C/L
REGEN C/L BOOST
OVER CURR DELAY

■ Acceleration Parameters

M1 FWD ACCEL
M2 FWD ACCEL
M1 REV ACCEL
M2 REV ACCEL
GEAR SOFTEN
SOFT START

■ Deceleration Parameters

M1 FWD DECEL
M2 FWD DECEL
M1 REV DECEL
M2 REV DECEL
EMG STOP DECEL
KEY OFF DECEL

■ Speed Parameters

M1 FWD MAX SPD
M1 FWD MIN SPD
M1 REV MAX SPD
M1 REV MIN SPD
M2 FWD MAX SPD
M2 FWD MIN SPD
M2 REV MAX SPD
M2 REV MIN SPD
CREEP SPD
PUSH SPD
MAX SPD VOLTS
EMG REV SPD
CON SPD ESCAPE

■ Throttle Parameters

THRTL TYPE
DIR SWITCHES
THRTL DEADBAND
THRTL RANGE
RAMP SHAPE

■ Output Parameters

PROG1 TYPE
PROG2 TYPE
BRK HOLD VOLTS

■ Fault Parameters

HIGH PEDAL CHECK
SRO CHECK
FAULT BEEP

■ Other Parameters

M1 IR COEFF
M2 IR COEFF
IR GRADE
BEEPER ALTER
SLEEP DELAY
MAIN CONT DELAY

■ COPY

COPY TO CONTROLLER
COPY TO PROGRAMMER

■ ABOUT

ABOUT CONTROLLER
ABOUT PROGRAMMER